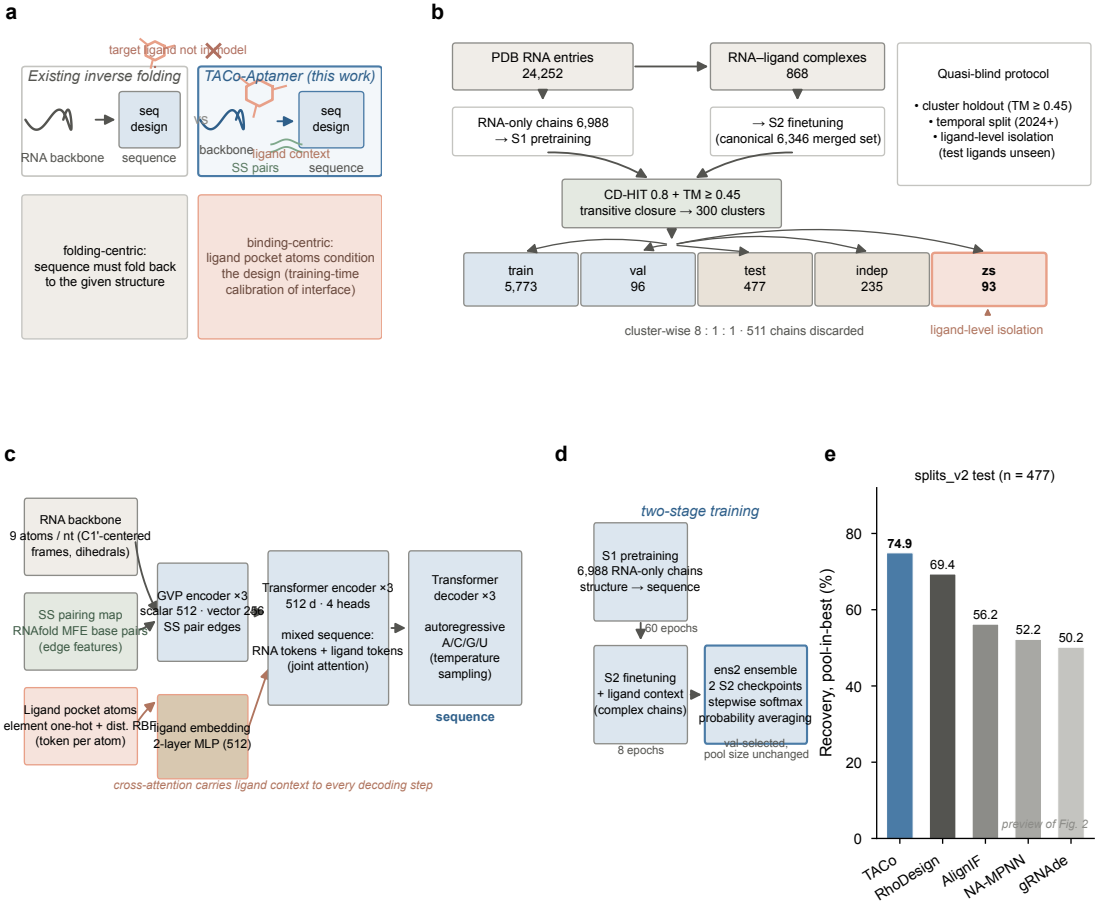

a
target ligand not in model
Existing inverse folding
seq design
RNA backbone
sequence
TACo-Aptamer (this work)
seq design
backbone
ligand context
SS pairs
sequence
folding-centric:
sequence must fold back
to the given structure
binding-centric:
ligand pocket atoms condition
the design (training-time
calibration of interface)
b
PDB RNA entries
24,252
RNA–ligand complexes
868
Quasi-blind protocol
• cluster holdout (TM ≥ 0.45)
• temporal split (2024+)
• ligand-level isolation
(test ligands unseen)
RNA-only chains 6,988
→ S1 pretraining
→ S2 finetuning
(canonical 6,346 merged set)
CD-HIT 0.8 + TM ≥ 0.45
transitive closure → 300 clusters
train
5,773
val
96
test
477
indep
235
zs
93
cluster-wise 8 : 1 : 1 · 511 chains discarded
ligand-level isolation
c
RNA backbone
9 atoms / nt (C1'-centered
frames, dihedrals)
SS pairing map
RNAfold MFE base pairs
(edge features)
Ligand pocket atoms
element one-hot + dist. RBF
(token per atom)
GVP encoder ×3
scalar 512 · vector 256
SS pair edges
ligand embedding
2-layer MLP (512)
Transformer encoder ×3
512 d · 4 heads
mixed sequence:
RNA tokens + ligand tokens
(joint attention)
Transformer
decoder ×3
autoregressive
A/C/G/U
(temperature
sampling)
sequence
cross-attention carries ligand context to every decoding step
d
two-stage training
S1 pretraining
6,988 RNA-only chains
structure → sequence
60 epochs
S2 finetuning
+ ligand context
(complex chains)
8 epochs
ens2 ensemble
2 S2 checkpoints
stepwise softmax
probability averaging
val-selected,
pool size unchanged
e
splits_v2 test (n = 477)
Recovery, pool-in-best (%)
0
20
40
60
80
74.9
69.4
56.2
52.2
50.2
TACo
RhoDesign
AlignIF
NA-MPNN
gRNAde
preview of Fig. 2